



SIMATS ENGINEERING

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Course Code : UBA0504

Slot : D

Course Name : DATABASE MANAGEMENT SYSTEM

Course Faculty: Dr. KESAVAN R

Project Title: SMART NATIONAL EXAM SLOT BOOKING AND RESULT MANAGEMENT SYSTEM

Module Photographs:



Project Description:

The **Smart National Exam Slot Booking and Result Management System** is a database-driven application designed to manage candidates, examinations, exam centers, slots, bookings, attempts, and results efficiently. The system is developed using **HTML, CSS, JavaScript, Bootstrap, Python Flask, and MySQL**. It provides a user-friendly interface while maintaining all important information in a structured relational database. Primary keys, foreign keys, constraints, and indexes are used to maintain data integrity and improve retrieval speed. The project demonstrates **file organization, indexing, and hashing** for efficient storage and searching of examination records. Hashing is used to demonstrate bucket-based record lookup and collision handling. Important SQL queries are analyzed using **EXPLAIN/EXPLAIN ANALYZE** to identify and improve query performance. The system uses **transactions** for exam-slot booking to ensure that seat availability, booking creation, and seat updates are completed safely. **COMMIT, ROLLBACK, and SAVEPOINT** are demonstrated to maintain database consistency. **Concurrency control** is implemented to prevent multiple candidates from booking the same available seat at the same time. The system also supports secure and consistent result processing. Overall, the project integrates important DBMS concepts with a web application to provide a reliable, efficient, and scalable examination slot booking and result management solution.

Student Signature

Guide Signature